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(2026-06-14, AnnikaStein)

PmgTopProcesses

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Regular meetings: Thursday, 15:00 (CERN time) - Shared time slot with Top Reconstruction / Top Plenary:

<https://indico.cern.ch/category/3332/>

Other PMG meetings: <https://indico.cern.ch/category/10312/>

Introduction

The PMG Top Processes subgroup is responsible for physics modelling aspects of the following processes:

- baseline samples for top processes (tt , single-top, $t(t)+X$)
- systematic uncertainty prescription for this baseline
- support in producing alternative predictions and associated uncertainties

Tasks included in its mandate are described in [PMGSubgroupMandates](#).

Old twiki: [PmgTopProcessesOutdated\(Run2\)](#)

Which sample to use for nominal and systematic?

All the information about existing samples, DSID and which ones are considered baseline and which one systematic are provided in the [central page](#). Matching this with the systematic table listed below should allow you to get the list of samples for each process.

Producing a new sample: ATLMCPROD JIRA

The conveners will be notified of updates on any ATLMCPROD JIRA ticket which contains the **Top** component. A summary of all issues is available in this [dashboard](#).

To open a ticket and request a new sample please follow the very detailed instructions given here: [MCrequest](#)

Note on derivation: Requesting the derivations is the requester job, that has also to post a link on the ATLMCPROD ticket.

For more details on how to modify the JOs and run tests locally or on the GRID please look at the ATLAS Top Workshop tutorial: [MCTutorial](#)

Higher Order Cross sections:

When you run on a sample it is necessary to normalise the sample to the right cross-section, if possible this is more accurate of the cross section obtained during the MC production (MC cross-section). The correction to the cross section is called k-factor, if a new calculation becomes available it is possible to just modify the k-factor without re-running the sample. The correct normalisation for each sample (or weight) needs to be evaluated as: MC cross-section * filter efficiency * k-factor/(Sum of all the weighted events) Here the weights can be the nominal or the alternative weight for systematic uncertainties/variations.

All the informations on MC cross-section, filter efficiency, k-factor are available through the [central page](#)

The list of higher order cross-sections employed is: [tt NNLO single top NNLO ttW](#)

The higher order cross-sections for t(t)X processes at 13.6 [TeV](#) are still being evaluated.

Systematic uncertainties

General recipes

PMG recommendations for the evaluation of systematic uncertainties can be found in [PmgSystematicUncertaintyRecipes](#). For uncertainties implemented as on-the-fly weights, PMG provides the [PMGSystematicsTool](#) for an automatic calculation and combination of the relevant uncertainties with correlations in ROOT or Rivet-based analyses: [Link Talk](#) [Link Git](#)

Always compare samples using the same type of generation: AF2/3 samples compared among them and Full Sim samples compared among them.

In addition to the uncertainty listed here also normalisation uncertainties need to be considered, based on the xs section used to normalise the sample. These can be derived in the Higher Order Cross sections pages.

Process-specific recipes

In the sections below you will find direct links and information for modelling uncertainties in top processes. The Top Processes group supports these uncertainty prescriptions, but one should be aware that uncertainties for

measurements may need to be more precise and additional studies may be needed to assess the need to add new uncertainties/modify the existing prescriptions.

We support also the usage of Sherpa as tt nominal for searches working in the high jet multiplicity or high pT regions, the samples exist in MC16 and produce them in MC23 is an open task. The systematic uncertainty receipt in this case is the same of the ttW/ttZ Sherpa samples.

ttbar, Single Top, ttH

Uncertainty	Type	Description	Additional comments	Key (see below)
Nominal	-	Powheg + Pythia 8	-	
Matrix element (NLO matching)	2 point	Compare with sample generated with pythia8 pthard = 1	This parameter regulates the definition of the vetoed region of the showering, important to avoid hole/overlap in the phase space filled by POWHEG and PYTHIA. This recommendation follows the description included in SciPostPhys.12.1.010	
Parton shower	2 point	Compare with Herwig 7	Samples with 7.1.3 and 7.2.*. Analysers should be using the newest version where possible in their analysis, or request additional samples	
Scale	Reweight	see the official PMG recommendation		
ISR alpha_s	Reweight	A14 Var3c, vary ISR alpha_s	Use weights "Var3cUp" and "Var3cDown"	
FSR	Reweight/direct variations	Vary FSR scale (ie FSR alpha_s)	Use weight "isr:muRfac=1.0_fsr:muRfac=2.0" and "isr:muRfac=1.0_fsr:muRfac=0.5". The variation with reduced scales is affected by large statistical fluctuations due to a large spread of weights, use the samples with direct variations in case you are limited from this effect.	
PDF	Reweight	NNPDF and PDF4LHC variations possible	Use event weights -- see this twiki for more info	
hdamp variation	2 point	Compare nominal with hdamp = 3.0 m_top	This uncertainty has to be applied also to tW samples (if generated with dynamic scales ATLMCPROD-10313), not to other single top production channels.	
top pt shape	2 point	(Only tt) Compare nominal with the nominal reweighted to NNLO		

recoil to top	2 point	(Only tt and t-channel) Compare nominal (recoil to color) with the alternative sample obtained with the recoil to top sample		
Interference between tt and tW	2 point	(Only tW) Compare nominal DR with the alternative DS sample		
Underlying Event	-	(Only tt) Pythia8 A14 Var1	Bulk samples available	
b-quark fragmentation	-	(Only tt) Pythia8 A14-rb		
top line shape	2 point	(Only tt) Compare POWHEG vs MADSPIN line shape		 
Colour reconnection	-	Compare with CR1 and CR2	(Only tt) The comparison must be done wrt CR0, not with the nominal. The recommendation is to take the largest variation.	 
Top mass	-	(Only tt) Compare nominal with the alternative samples obtained with different mass points	This systematic variation will only be relevant in specific analysis that target top-mass sensitive observables	

Comments relating to uncertainties

Key	Information
	See section below relating to this specific uncertainty
	These uncertainties as for precision top measurements, as top mass, or other measurements that expect to be sensitive to these effects.
	Careful consideration should be given to whether an analysis should be sensitive to a variation on hdamp which will impact the modelling of the first emission. In some analyses away from the "bulk" phase-space, this uncertainty should not need to be considered.
	It should be noted that by probing shower scale effects using automated shower weights includes an NLO compensation term which would not be present if generating a standalone sample with these scales. This will lead to smaller uncertainty than would be estimated with the other method. However, it is accepted that scale variations between a factor of 2 and sqrt(2) without an NLO compensation term will be similar to a weighted variation factor of 2 with an NLO compensation term, so we are not concerned that this uncertainty is therefore underestimated using the shower weights included in these uncertainties.
	Not all analyses will be affected by this uncertainty, for example analyses measuring the ttbar differential xs shouldn't be affected. Other analyses sensitive to the shape of the top (or lepton, jet pt) to extract a parameter (e.g. top mass, ttbar xs) could be affected by the known mismodelling of this quantity and need to check the effect of including this additional uncertainty. In the past the mismodelling of the top pt was covered (by chance) by the difference between aMcAtNlo+Pythia8 and Powheg+Pythia8, but with the improvement of the matching uncertainty definition this is not the case anymore. Instructions on how to perform the reweighting are given below.

ttbar+HF (tt + ≥ 1 b-jet and tt+ ≥ 1 c) - preliminary

The analyses sensitive to the modelling of tt + Heavy Flavour jets (HF) should use a more robust modelling systematic model than the one for inclusive ttbar. The exact recommendation depends on the analysis details and must be discussed with PMG conveners at the initial stage of the analysis. In general, the systematic model should be similar to the systematic models used in past measurements, e.g. tt+c (<https://cds.cern.ch/record/2789804> [arXiv:2409.11305](#)) or 4tops 1LOS (<https://cds.cern.ch/record/2722655> [arXiv:2106.11683](#)) or ttH(bb) Legacy (<https://cds.cern.ch/record/2790108> [arXiv:2407.10904](#)). The general recommendation is:

1. Divide the inclusive ttbar physics process into categories based on the number of additional heavy-flavour jets.

The three main categories are:

- tt+ ≥ 1 b - This category should be obtained from the Powheg+Pythia8 simulation of the tt+bb process using 4FS (in mc20: 603003,603190,603192; in mc23: 601226-601228).
- tt+ ≥ 1 c - This category should be obtained from the Powheg+Pythia8 simulation of the tt process in 5FS (in mc20: 410470-410472; in mc23: 601229-601230). A particle-level selection should be applied on the number of jets containing B-hadrons and D-hadrons (see the above references).
- tt+light - The same approach as for tt+ ≥ 1 c category, just using different particle-level criterion.

Comment 1: Further categorization of tt+ ≥ 1 b and/or tt+ ≥ 1 c into subcategories may be necessary, depending on the analysis details, see for example the ttH(bb) Legacy measurement.

Comment 2: For single-lepton selection of events, ~50% of ttbar events will have a c-jet from the W-boson decay. This is a potential challenge when categorizing tt+ ≥ 1 c. In this case, contact the conveners.

2. For each category, use the systematic uncertainties as described in the section for inclusive ttbar recommendations (ptHard1, Herwig7, scale, ISR, FSR variations). Additionally, for tt+ ≥ 1 b, the 2-point systematic dipole vs global recoil is recommended, as it was demonstrated in the ttH(bb) Legacy measurement. For the 2-point tt+ ≥ 1 b systematics, use the alternative tt+bb 4FS samples.

3. In some cases of past analyses, the so-called "4FS vs 5FS" uncertainty was included. It was obtained as a 2-

point systematics comparing two samples: $t\bar{t}+b\bar{b}$ in 4FS and inclusive $t\bar{t}+b\bar{b}$ in 5FS (with particle-level selection of $t\bar{t} \rightarrow \geq 1b\text{-jet}$). This systematic convolutes several effects (most notably a ME vs PS calculation of the additional b -jets) and is not expected to provide a good estimate of the 5FS vs 4FS uncertainty, therefore it is not recommended.

$t\bar{t}Z$, $t\bar{t}t\bar{t}$, $t\bar{t}g\gamma$

These processes are generally not in Powheg+Pythia, where we have the most detailed set of uncertainties. The main idea is that we should cover all aspects of the event generation: matching, scale choice, showering algorithm, shower scales, PDF also for these rare processes.

For $t\bar{t}Z$, $t\bar{t}t\bar{t}$, $t\bar{t}g\gamma$:

Uncertainty	Type	Description	Additional comments
Nominal	-	aMcAtNlo + Pythia 8	-
Matrix element	2 point	Sherpa if available	Typically difficult to find additional generators providing same processes
Parton shower	2 point	aMcAtNlo + Herwig 7	May be missing for some $t\bar{t}V$ processes
Scale	Reweight	see the official PMG recommendation	
Shower variations in Showering	A14 Var3c	Dedicated samples with the A14 Var3c variations	Availability of these sample can be limited
PDF	Reweight	ME PDF variations should be available	Using event weights -- see this twiki for more info

$t\bar{t}W$

If the nominal includes the EW corrections through reweighting, these do not need to be also added in the variations. The important aspect is to be consistent, if the EW corrections are off in the variations also turn them off in the nominal

Uncertainty	Type	Description	Additional comments
Nominal	-	Sherpa	-
Matrix element	2 point	-	Discussion ongoing if this uncertainty is really needed or can be replaced with internal variations of some parameter
Parton shower	2 point	Powheg + Herwig 7 vs Powheg + Pythia8	
Scale	Reweight	see the official PMG recommendation	Using event weights, Sherpa allows to do a coherent variations of the scales in the matrix element and the showering
Hadronization	2 point	Lund vs Cluster (default)	Dedicated samples with varied hadronization model
PDF	Reweight	ME PDF variations should be available	Using event weights -- see this twiki for more info
Matching Scale	muQ	Dedicated samples with varied scales (QSF)	

Precision recommendations for $tt+W$

The ttW process has been studied in the context of $ttH(ML)$ and ttW measurements, where it's been found that electro-weak corrections are non-negligible. Several theoretical progress have been on the calculation to improve the precision of the sample and the agreement with data. A NLO sample up to 2 additional jet emissions, including these EW contributions, using the [FxFx](#) algorithm has been made in [FxFx EWK jets](#). The most precise theory calculation is [aNNLO](#).

Even with these precise recommendations it is not trivial to normalise the sample because separate NLO EWK components can be added and the **normalisation** depends on the samples used. We recommend 2 approaches, depending on the analysis.

- ttW is the major background or the signal:
 - 2 separate samples: ttW (QCD+EWK lead) and ttW (EWK-sub)
 - Normalise the first to (QCD+EWK lead) from aNNLO and the second to EWK-sub from [FxFx](#).
- ttW is a small background:
 - Just take $ttW(QCD)$ sample and apply (EWK-lead+EWK-sub) normalisation correction

See also this [presentation](#) for the case-by-case normalisation proposed for Run2 analyses.

Sherpa allows the use of different MC weights to add the EW contributions. [This AQP](#) details the usage of these weights and maps them to the diagrams that are added by each of them; it also explains in detail the internal variations (see table above) available in Sherpa that should be used if ttW is your signal or a major background.

These EW weights may be added using two different schemes, additive or multiplicative, and their impact has been studied [here](#) (Appendix L). The difference was found to be small so the additive scheme was used for the paper.

For Sherpa, the EW-lead contributions can be added via reweighting, using the *ASSEWLO1 MC weight (but keeping the sum of weights of the nominal MC weight). Since Run3 is currently lacking a dedicated ttW EWK-sub sample, its contribution can be estimated using the Sherpa internal weight *ASSEWLO1LO2LO3 and normalised to the aNNLO prediction.

ttGamma

These processes need to be evaluated combining an NLO sample where the gamma is produced during production and a LO sample where the photon is emitted in the decay. Uncertainties need to be evaluated for each sample. Inclusive samples (as NLO tt) also includes the radiation of gluons performed by the showering algorithm, to avoid double counting a specific technique needs to be applied. Details are explained [Section3](#).

Lineshape uncertainty

All our ttbar generators, excluding bb4l and NLO-dec (under development), consider the top-quarks on shell in production and then include approximations to recover the shape of the top mass with a proper width and high off-shell behaviours. This makes, especially the highly off-shell region, unreliable. A perfect solution would be to use bb4l, where no approximations to the lineshape are employed. The lineshape uncertainty targets this ambiguity of our "standard" generators, aiming to compare the 2 existing implementations of the smearing and off-shell corrections: Powheg and Madspin.

Studies show that the agreement between our nominal and bb4l is good in the region around the top-quark mass, see [bb4l PUB note](#), so we recommend this uncertainty only for analyses targeting the tails of the top mass or mttbar distribution. The discussion on this uncertainty is still open so the conclusion and application suggestions may change.

Color Reconnection

Pythia8 provides a number of [colour reconnection models](#). Studies were performed in ATLAS where three new models were tested by taking A14, and switching to CR0, CR1 or CR2 and then performing a returning of parameters applicable to colour reconnection (see [ATL-PHYS-PUB-2017-008](#)). Analyses which need to estimate the impact of changing the colour-reconnection model should take the largest of these variations as uncertainty. If the two variations go in opposite direction, the analysis appears to be very sensitive to this variation and taking the largest doesn't guarantee the most conservative approach (e.g. fit) analyses should consider both variations.

A new sample has been generated with an additional variation in the color reconnection scheme (ERD On samples). While the default Pythia8 settings do not allow to the decay products to reconnect in these samples the reconnection is enabled and the range for reconnection is also enlarged. These samples are more similar to Herwig7 setting, compared to the nominal Pythia8. These samples are currently available only for testing purpose, they are not included in any recommendation.

Recoil to Top uncertainty

The modelling of the parton shower of the b-quarks in $t \rightarrow W b$ is affected by the gluon recoil scheme. The default Run2 samples employs the recoiling against b-quarks (recoilToColor=ON in the pythia8 settings), independently on the color connection of the emitted gluon. This setting changes the modelling of second and subsequent gluon emission from quarks produced by coloured resonance decays, such as the b-quark in a $t \rightarrow W b$ process (but it has no impact for example on $Z \rightarrow b\bar{b}$ decays), practically deciding how the momentum is re-arranged between the W and the b. Pythia8 developer suggested that the recoilToColor scheme cut large-angle radiation off the b-quark and a better scheme would be the recoilToTop, where the top-quark itself works as recoiler for the second and subsequent emissions. This option give wider-angle gluon radiation, resulting in energy deposits that do not get clustered into the b-jet, and lower gluon-energy emission, altering the modelling of the b-quark fragmentation and hardening the b-hadron momenta. This text is largely taken by the first ATLAS analysis using this recommendation: [SMT mass measurement](#).

The recoil to top scheme should, currently, be taken as an uncertainty with respect to the nominal recoil scheme. This uncertainty is currently enlarged by the lack of a dedicated tuning that takes into account the recoilToTop scheme. The recoilToTop option is implemented in a dedicated userhook provided by the Pythia8 authors to ATLAS in Pythia versions < 8.309 and available in more recent versions changing the parameter ([TimeShower:recoilStrategyRF](#))

Please note that these variations are only available for the $t\bar{t}$ and t-channel process. The t-channel sample is mainly used by the t-channel top-mass analysis. If other top-processes are considered for measuring the top-quark mass please get in contact as some additional samples for other processes may be needed.

Reweighting top-pair events to NNLO prediction

There are some issues relating to top modelling which are inextricably linked to the precision of our MC calculations (eg top pT mismodelling). Higher order corrections improves this mismodelling, but reweight to these predictions is complicated and using only one observables (e.g. top pT) will typically result in destroying the precision of other differential variables due to cross-correlations. Using a recursive approach to the reweighting, including multiple variables, partially solves this issue. Tools for the reweighting have been developed by member of the top group:

- [TTBarNNLOReweighter](#) (Original) package - <https://gitlab.cern.ch/lserkin/TTbarNNLOReweighter>
- [TTBarNNLOReweighter](#) (Recursive) package - <https://gitlab.cern.ch/pinamont/TTbarNNLOReweighter>

They can also be adapted to take as input a custom higher order calculation obtained with Matrix or [HighTEA](#), this is useful for 13.6 energies.

It is possible to also call this tool inside a rivet routine, more information and instructions on how to do so are shown here: [J.Jamieson](#)

Searches may find useful to use the reweighted sample as nominal, in this case all the uncertainties are to be evaluated coherently, an example of this kind of approach can be find here: [tt resonance](#)

$t\bar{t}/tW/WWbb$ Interference

tW diagrams at NLO QCD clearly overlap with LO $t\bar{t}$ diagrams. To account for this overlap, the tW process is generated removing these diagrams to allow a merging which does not double-count, but it is possible to be in a region of phase-space which becomes sensitive to the interference between these processes in the resonant region where $t+Wb$ is produced with $m(Wb) = m(\text{top})$. The nominal Powheg+Pythia8 tW sample uses the diagram removal scheme (DR) which removes the overlap at the cross-section level and keeps the interference in the prediction. Alternative treatments are possible on the interference, the current recommendation is the comparison with DS. In case an analysis would like to employ DS as nominal please contact us. The usage of dynamic scales in general reduces this uncertainty.

The correct approach to include the interference, that also entirely remove this uncertainty, would be to consider the $t\bar{t}$ and tW as a single process and so having a simulation of [WbWb](#). We have currently a sample produced using bb4l that simulate the full [WbWb](#) process. The dileptonic sample is currently available and the semileptonic one will be ready soon (together with a newer version for the dileptonic). A complete set of uncertainties around this sample is being developed in one of the ongoing projects.

Ongoing Qualification Tasks and Open projects

Here there is the list of all open and ongoing projects and QT tasks. If the projects are described in a JIRA ticket they are directly taken from the list of AGENE tickets with the label "Top".

To get started to collaborate on a MC project look at recent tutorial [MCTutorial](#)

If you are interested in all PMG related ongoing projects, look at [GIT twiki page](#)

Open/Ongoing Projects

All the top-processes specific projects can be converted in service tasks for your analyses, please contact the [TopProcesses](#) conveners and we will open a formal JIRA to monitor the progresses.

[Open/Ongoin Project Google Docs](#)

Below are included the list of general GIT open projects. This also include projects that are not limited to top but could be extremely helpful also for top physics.

[GIT Kanban board](#)

Completed Projects

Known modelling issues

Please feel free to populate this table with any known issues relating to samples in this subgroup as it may help others in the future.

Generator	Summary	DSIDs	Issue	Date added
-	No tau spin correlation	-	These Sherpa samples lack spin correlation in the decay products of taus. There is difficulty propagating spin information from W boson to the tau to its final state (see https://gitlab.com/sherpa-team/sherpa/issues/15). This affects ttbar samples.	27 Jan 2020 12:27
Pythia8	Bug in colour flow in top decay dipoles	500460, 500461, 500462, 500463, 500800, 502957, 502958, 504554, 600027, 600028, 600031, 600639, 600641	This bug was reported by P. Skands as being introduced in 8.240 and fixed in 8.245. It was also introduced to Pythia 8.3 in the early days, and was fixed with version 8.303. The size of this bug is not known. The list of DSID is non-exhaustive at this stage, only showing R21 productions.	04 Mar 2021 15:59

Practical information

Rivet Routines for MC Validation

Check the instructions from [RivetAtlas](#) the to see how to get started with Rivet.

ATLAS_2012_I1094568	Measurement of ttbar production with a veto on additional central jet activity	7 TeV
ATLAS_2013_I1243871	Measurement of jet shapes in top quark pair events	7 TeV
ATLAS_2014_I1304688	Measurement of jet multiplicity and transverse momentum spectra in top events	7 TeV
ATLAS_2015_I1345452	Pseudo-top-antitop cross sections	7 TeV
ATLAS_2015_I1397637	Boosted ttbar differential cross-section	8 TeV
ATLAS_2015_I1404878	ttbar (to l+jets) differential cross sections	8 TeV
ATLAS_2015_I1376945	Colour flow in ttbar lepton+jet events	8 TeV
ATLAS_2015_I1390114	$\$tt+b(b)\$$	8 TeV
ATLAS_2017_I1626105	tt differential cross-section in dilepton channel	8 TeV
ATLAS_2017_I1495243	tt dilepton gap-fraction	13 TeV
ATLAS_2018_I1646686	tt differential cross-section in all-hadronic channel	13 TeV
ATLAS_2018_I1656578	tt + jets differential cross-section in lepton+jets channel	13 TeV
ATLAS_2018_I1705857	$tt+bb$ fiducial and differential cross-section	13 TeV
ATLAS_2018_I1677498	WWbb differential cross-section	13 TeV
ATLAS_2019_I1724098	measurement of large-R jet shape (light,W,t) in ttbar and singletoop (36 ifb)	13 TeV
ATLAS_2019_I1759875	lepton kinematic in dilepton ttbar (36 ifb)	13 TeV
ATLAS_2019_I1750330	tt differential cross-section in lepton+jets (36 ifb)	13 TeV
atlasphys-top/reco/RivetRoutines ↗	Additional rivet routines from the Top WG (including MC/MC and unvalidated routines)	Various
pmg/rivet-routines ↗	On-going rivet routines from ATLAS	Various

Public results

Public results from the Top group can be found in this [link](#)

Major updates:

-- [FrankSiegert](#) - 2019-04-05

Responsible: Marcos Miralles and Peter Berta

Last reviewed by: **Never reviewed**

Topic revision: r153 - 2026-06-14 - [AnnikaStein](#)

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